EduGenie

Project Design Phase

Software Project Design Document

Prepared for Academic Project / Innovation Activity

1. Introduction

The Project Design Phase converts the approved requirements of EduGenie into a structured technical design. It defines the system architecture, modules, data flow, user interface, database structure, and major technologies required to develop the application.

2. Design Objectives

The design aims to create a simple, reliable, scalable, and student-friendly educational platform. The system should separate major functions into manageable modules so that development, testing, maintenance, and future improvements can be carried out efficiently.

3. System Architecture

EduGenie can follow a layered architecture consisting of a presentation layer, application or business-logic layer, data layer, and optional AI/ML service layer. The presentation layer handles user interaction, the application layer processes requests, the data layer stores information, and AI/ML services can provide intelligent educational assistance.

4. Major System Modules

- User Management Module – registration, login, profile, and access control.
- Learning Module – subject-wise learning content and explanations.
- Doubt Assistance Module – helps students clarify academic questions.
- Study Planner Module – creates and manages study schedules.
- Quiz Module – provides practice questions and evaluates responses.
- Progress Tracking Module – records learning activity and performance.
- Resource Module – organizes useful educational materials.
- Administration Module – manages users, content, and system settings.

5. User Interface Design

The interface should provide clear navigation and simple screens for login, dashboard, subjects, doubt clarification, study planning, quizzes, resources, and progress tracking. The design should be responsive so that it can be used on suitable computers and mobile devices.

6. Database Design

Entity	Example Data
Student	Student ID, name, email, preferences
Subjects	Subject ID, subject name, description

Study Plan	Plan ID, student ID, tasks, dates, status
Quiz	Quiz ID, subject, questions, answers
Progress	Student ID, activity, score, completion status
Resources	Resource ID, title, subject, resource type

7. Data Flow

A typical interaction begins when the student enters a request through the interface. The application processes the request, retrieves required information from the database or relevant service, and returns the result to the student. Learning activities such as quiz attempts and progress updates can also be stored for later use.

8. Technology Design

- Front end: HTML/CSS/JavaScript or a suitable web/mobile framework.
- Back end: Python, Java, Node.js, or another suitable server-side technology.
- Database: MySQL, PostgreSQL, MongoDB, or another suitable database.
- AI/ML: Optional Python-based machine learning or AI services for intelligent features.
- Development tools: A suitable IDE, version control system, and testing tools.

9. Security Design

The system should use appropriate authentication and authorization mechanisms. User data should be protected through secure storage and controlled access. Sensitive information should not be exposed unnecessarily, and regular backups should be considered.

10. Testing Considerations

Each module should be tested individually and then tested together as an integrated system. Functional testing, usability testing, performance testing, security checks, and user acceptance testing can be included before final deployment.

11. Future Enhancements

Future versions may include multilingual learning support, advanced personalization, improved analytics, teacher dashboards, additional AI-based learning tools, and integration with other educational platforms.

12. Conclusion

The Project Design Phase provides the technical blueprint for EduGenie. A clear design helps the development team implement the requirements systematically and provides a strong

foundation for testing, deployment, and future enhancement.